



An intelligent measurement method for the particle distribution of open-pit rock piles with fuzzy boundaries

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ABSTRACT

After rock blasting, the measurement of rock fragment particle size distribution (PSD) is a critical step in aggregate size control and blast process optimization. However, in open-pit scenarios, overlapping and occlusion between rock fragments, as well as lighting and shadow effects, lead to blurry edges, significantly affecting the accuracy of existing rock segmentation models, thereby increasing the bias in PSD estimation. This study proposes a novel instance segmentation model, CTURNet, which integrates a Multi-Level Contour Enhancement Module (MLCEM) to improve the contour features of blurry rocks. Additionally, a Transformer-based multi-scale longitudinal fusion feature pyramid structure is designed to enhance adaptability in complex scenes. Furthermore, a Umask segmentation head is incorporated into the mask branch, utilizing a Unet structure to improve the mask segmentation accuracy of rocks with blurry edges. Finally, comparative experiments with various advanced methods are conducted, and the model is applied in real-world scenarios. Experimental results show that the proposed method achieves optimal performance, with AP and IoU improved by 3.29% and 4.34%, respectively, compared with the baseline model, demonstrating excellent accuracy and stability.

1. Introduction

Rock extraction is a complex and multi-stage process, encompassing several stages from initial blasting to subsequent processing. Among these, rock blasting is the most efficient fragmentation method, as it not only determines the preliminary fragmentation of the rock but also directly influences subsequent processes such as crushing, screening, and transportation [1–3]. Therefore, to achieve optimal blasting results, it is essential to design the blasting pattern rationally to obtain the best fragment size. Moreover, there is a trade-off between rock fragment size and economic benefits; larger fragments require secondary breaking [4] to reduce the size of boulders, while smaller fragments increase mining costs [5]. Thus, in-depth research on the size distribution of rock fragments after blasting is particularly important [6,7]. This research not only helps optimize blasting parameters but also effectively reduces energy consumption and improves overall production efficiency, carrying significant engineering and economic value.

Currently, the commonly used methods for measuring rock size are manual sieving and computer image detection. The manual sieving method involves high labor intensity, is time-consuming, and is difficult to obtain stable and reliable rock fragmentation information [8]. In contrast, computer vision technology can quickly collect data from

the blasting site, greatly enhancing the automation and intelligence of rock particle size analysis. However, traditional image processing algorithms rely on edge detection [9–11], which is limited to cases where the rock fragments in the image have clear definitions and boundaries. These methods struggle when dealing with more complex contours. Fortunately, with the significant achievements of Convolutional Neural Networks (CNNs) in tasks such as image classification, object detection [12,13], and image segmentation [14,15], CNNs have gradually been applied to the field of rock fragmentation analysis. Object detection and image segmentation are considered more accurate methods for rock particle size identification. Object detection algorithms can identify the position and quantity of rocks but struggle to precisely capture the contours of the rocks and cannot directly calculate their size and volume. In contrast, image segmentation methods can accurately separate individual rocks from the image and directly predict their two-dimensional size, making it the mainstream method for rock particle size measurement. However, in open-pit scenarios, where rocks have similar colors, severe stacking, and fuzzy boundaries, the challenge of accurately measuring the rocks increases. Existing image segmentation methods still face challenges such as limited capability to

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extract rock contour features and insufficient integration of global and local information, which to some extent restricts the improvement of segmentation accuracy.

To address the aforementioned issues, this paper designs a novel instance segmentation network, CTURNet, aimed at improving the instance segmentation accuracy of rocks with fuzzy boundaries in open-pit environments. The algorithm is based on a two-stage segmentation network and innovatively incorporates a series of network modules, including an enhanced rock edge feature module, a global information fusion module, and a multi-scale fully convolutional pixel classifier, to achieve high-precision rock segmentation. The main contributions of this paper are as follows:

(1) A multi-level contour enhancement module (MLCEM) is designed. This module uses convolutional kernels to simulate the Sobel operator and performs multiple contour enhancement operations on the feature maps, thereby enhancing the backbone network's ability to extract blurred contour features.

(2) A Transformer-based Feature Pyramid Network (TF-FPN) is proposed. This module adds a Transformer information fusion branch to the traditional FPN architecture, enabling it to simultaneously process both local details and global semantic information, effectively addressing the long-distance dependency issues between objects in complex stacked scenes.

(3) An UMask module is designed. This module combines the encoder-decoder architecture of U-Net to provide multi-scale information for the mask branch, improving the segmentation accuracy of rocks with similar colors.

(4) An effective instance segmentation network, CTURNet, is proposed for open-pit rocks with fuzzy boundaries. Comparisons with advanced algorithms demonstrate that CTURNet outperforms other models in both quantitative and visual evaluations. In practical PSD measurement projects, it meets the actual accuracy requirements.

The remainder of this paper is organized as follows: Section 2 reviews the literature related to rock pile particle size distribution measurement and provides a summary and discussion. Section 3 presents the proposed method and the design of the network modules in detail. Section 4 conducts comparative experiments with multiple models based on engineering practice and calculates the particle size distribution (PSD). Section 5 summarizes the engineering significance, robustness, and advantages and disadvantages of the proposed method, while Section 6 provides the conclusion and outlines future research directions.

2. Related works

In recent years, the application of artificial intelligence in mining engineering has made significant progress, particularly in rock breaking, blasting optimization, rock movement prediction, and intelligent mine management. Machine learning and deep learning technologies have become key enablers driving the advancement of intelligent mining. For example, Nguyen et al. [16] used a convolutional neural network (CNN) to automatically predict the velocity and direction of rock fragments during blasting; Huang et al. [17] developed a blasting vibration prediction model based on a support vector machine (SVM); and Monjezi et al. [18] combined a random forest with a genetic algorithm to optimize blasting parameters, effectively improving rock fragmentation and reducing overbreak rates.

In the mining production process, accurate identification of rock fragmentation characteristics and measurement of particle size distribution are crucial for evaluating blasting effectiveness and optimizing production parameters. Therefore, image-based particle size analysis has gradually become an important application direction of artificial intelligence in the mining field. Existing methods can be broadly divided into two categories [19], one category is pattern recognition-based methods, which construct neural network models to extract and analyze

image features, thereby establishing a mapping between rock size distribution levels and image features for the prediction and evaluation of PSD. The other category is image segmentation-based methods, which use segmentation algorithms to accurately segment rock particles in images, measure pixel information of each particle, and calculate the overall PSD based on actual scale mappings [14].

In pattern recognition methods, researchers typically employ techniques such as Fourier transform [19], wavelet transform [20], and filtering algorithms [21] to extract rock features in images, and then input the extracted features into models such as neural networks [22] or support vector machines (SVM) [23,24] to model the rock PSD. These methods do not classify images on a pixel-wise basis but instead infer the size distribution based on holistic image features. The performance of such approaches heavily depends on the quality and diversity of the training data. However, for open-pit rock scenes, the large morphological variability of rock piles may result in insufficiently representative samples within the training dataset. This leads to difficulties in accurately identifying rock granularity in environments with significant data distribution shifts, and may cause considerable detection errors.

Compared with pattern recognition methods, image segmentation methods offer better interpretability. Image segmentation includes traditional segmentation algorithms and deep learning-based algorithms, both of which can be applied to rock fragmentation segmentation tasks. Traditional segmentation methods include edge detection [25], morphological processing [26], and watershed segmentation [27–29]. These methods usually require high image contrast to precisely extract closed rock boundaries, and thus place stringent demands on the image acquisition environment [14]. Additionally, although commercial software based on traditional image processing algorithms, such as WipFrag and Split-Desktop [30], is widely used for rock segmentation tasks, these tools often suffer from inaccurate recognition in practice and typically rely on manual correction, introducing subjectivity into the results [31]. In general, traditional image processing methods are highly dependent on imaging conditions, involve complex data processing, and some edge detection algorithms tend to generate “false edges”, which compromises segmentation accuracy and overall detection performance.

Among deep learning-based image segmentation methods, semantic segmentation and instance segmentation are the main approaches. Semantic segmentation classifies each pixel in the image and assigns a category label to each, whereas instance segmentation further distinguishes between different instances of the same object class, providing pixel-level object segmentation. Most researchers currently focus on semantic segmentation models, particularly fully convolutional network (FCN)-based models, which have rapidly developed and are widely applied in fields such as medical imaging [32], remote sensing [33], and autonomous driving [34]. In rock segmentation tasks, these methods also demonstrate strong adaptability and accuracy. Mainstream FCN semantic segmentation networks include U-Net [35], DeepLab [36], and SegNet [37], which generally adopt an encoder-decoder architecture to progressively extract image features and perform pixel classification, offering strong feature representation and reconstruction capabilities. For example, Changyu Jin [8] proposed U-CARFNet, a network incorporating attention mechanisms and residual learning, for fine-grained segmentation of rock pile images; Qinpeng Guo [38] proposed FRRnet+ based on the U-Net model, combining it with a watershed algorithm to effectively separate segmented rock piles into individual rocks; Xiaoxiong Zhou [14] proposed MSD-Unet with a dual U-Net architecture for real-time analysis of muck in TBM tunnel excavation. These methods fall under semantic segmentation, which, although capable of pixel-level classification, still requires post-processing methods such as watershed to separate individual rock fragments.

Instance segmentation, on the other hand, offers an end-to-end approach for directly predicting individual rock blocks, enabling precise segmentation and labeling of each object in the image. Leading instance

segmentation networks include YOLACT [39] and, R-CNN [40], and other models. In rock pile segmentation tasks, Weidong Qiao [41] combined object detection and semantic segmentation networks to design a novel architecture, demonstrating the effectiveness of instance segmentation for rocks; Vu. T. [42] applied Mask R-CNN to train on rock images captured at actual blasting sites in open-pit mines; Mingze Li [31] proposed a hybrid model called RDT-FragNet, using deformable convolution and Transformer attention to improve rock recognition accuracy. Although these studies achieved favorable instance segmentation results, they still lack effective contour enhancement and boundary segmentation algorithms for handling blurry rock boundaries and complex rock shapes.

Instance segmentation is an end-to-end approach that directly predicts individual rock blocks, enabling precise segmentation and labeling of each object in the image. Leading instance segmentation networks include YOLACT [39] and Mask R-CNN [40]. In rock pile segmentation tasks, Weidong Qiao et al. [41] combined object detection and semantic segmentation networks to design a novel architecture, demonstrating the effectiveness of rock instance segmentation. Vu. T. et al. [42] trained a Mask R-CNN model on rock images captured from actual open-pit blasting sites. With the success of the Transformer in natural language processing, its powerful feature modeling capability has been increasingly applied to image segmentation tasks, achieving remarkable progress. Mingze Li et al. [31] proposed a hybrid model, RDT-FragNet, for rock segmentation, which improved rock recognition accuracy using deformable convolution and Transformer attention mechanisms. Cheng et al. proposed Mask2Former [43], a unified segmentation framework applicable to semantic, instance, and panoptic segmentation, achieving superior performance across multiple datasets. Wang et al. [44] introduced Pyramid Vision Transformer V2, which enhanced multi-scale feature modeling through a hierarchical attention mechanism. Ma et al. [45] improved the fine-grained segmentation performance of Swin Transformer via adaptive feature reallocation. Although these studies achieved favorable instance segmentation results, they still lack effective contour enhancement and boundary segmentation algorithms for handling blurry rock boundaries and complex rock shapes.

This paper proposes the CTURNet network, which combines the MLCem module and the UMask module to effectively improve rock segmentation capability under fuzzy boundaries. By introducing the TF-FPN network, global correlations between multi-scale features are enhanced, significantly improving rock segmentation accuracy. Additionally, the accuracy of open-pit rock pile particle size distribution measurement has been effectively improved.

3. Proposed method

3.1. Network architecture

This paper presents a novel two-stage instance segmentation network, CTURNet, which can accurately identify and segment post-blast open-pit rock fragments. Fig. 1 shows the complete network architecture. CTURNet consists of several modules, including the ResNet backbone network with the integrated multi-level contour enhancement module, the FPN network with Transformer-based global feature extraction, the Region Proposal Network (RPN) for generating rock region candidate boxes, the ROI Align layer, the mask segmentation head integrated with the UMask network, the classification head, and the bounding box regression head.

An innovative contour enhancement network is incorporated into the CTURNet backbone, functioning as a multi-level enhancement architecture. This network effectively improves and strengthens the edge detail information of the rocks, providing a solid feature foundation for downstream tasks. In the multi-scale part, we designed TF-FPN, an FPN network integrated with a Transformer branch. This design not only leverages FPN to enhance multi-scale information processing capabilities but also enriches the feature map with global information

through the global feature extraction characteristics of the Transformer. In the pixel classification branch, we adopted a segmentation structure similar to U-Net instead of the traditional Mask branch, namely the UMask module. By incorporating multi-scale ROI information and a channel attention mechanism, along with a refined multi-scale feature fusion strategy, the UMask module not only ensures precise capture of local texture details but also significantly improves the segmentation performance of rocks with blurry edges, thus effectively enhancing the accuracy of rock particle size distribution detection. Next, we will provide a detailed introduction to the design ideas and implementation details of each module.

3.2. Multi level contour enhancement module

CNN networks exhibit a low-pass filtering effect on images, causing the high-frequency contour features to become blurred after processing [46]. Precise contour information is crucial for recognizing and distinguishing features of different rock instances. Due to the complex morphology and contour characteristics of rocks, we designed a Multi-Level Contour Enhancement Module (MLCEM). Unlike traditional single-layer Sobel-based edge enhancement methods, MLCEM introduces the Sobel operator into multi-scale feature layers and strengthens edge regions through cross-layer fusion and spatial attention. This helps the backbone better recognize and process contour features of rocks with fuzzy boundaries, thereby improving detection accuracy.

As shown in Fig. 2, the MLCEM structure progressively extracts image features through stacked 3×3 convolutional layers and 3×3 average pooling layers. After four stages of convolution and pooling operations, the contour representation capability of the feature maps is continuously enhanced. Along with the gradual blurring of the feature maps, the model is able to repeatedly focus on and extract edge features within the image.

$$F_{c,i+1} = \Phi_{3 \times 3}(\text{AP}_{i \neq 0}(F_{c,i})), (0 \leq i \leq 4, F_{c,0} = F_{\text{input}}) \quad (1)$$

where $F_{c,i} \in \mathbb{R}^{C \times H \times W}$, AP denotes a 3×3 average pooling operation, which is applied when $i \neq 0$. $\Phi_{3 \times 3}$ consists of a 3×3 convolutional kernel with padding of 1 and a stride of 1, followed by a Batch Normalization layer and a ReLU activation function.

After the aforementioned convolution and pooling operations, the Contour Enhancement Module (CEM) is applied to further strengthen the edge information within the feature maps. The purpose of the CEM is to enhance the edge features of the feature maps. As illustrated in Fig. 3, each feature map obtained from the convolutional layers is further processed by a CEM. The resulting four enhanced feature maps are then concatenated to form a more integrated feature representation, followed by a 1×1 convolution operation to adjust the number of channels. The computation is defined as follows:

$$F_{cem,i} = \text{CEM}(F_{c,i}), (0 \leq i \leq 4) \quad (2)$$

$$F_o = \Phi_{1 \times 1}(\text{Concat}[F_{cem,1}, F_{cem,2}, F_{cem,3}, F_{cem,4}]) \quad (3)$$

where $F_{cem,i} \in \mathbb{R}^{C \times H \times W}$ represents the contour-enhanced feature map, and $F_o \in \mathbb{R}^{C \times H \times W}$ denotes the fused feature map. $\Phi_{1 \times 1}$ includes a 1×1 convolution with padding of 1 and stride of 1, followed by a batch normalization layer and a ReLU activation function.

Finally, the sigmoid function is applied to F_o to compute the weight map, where the edge contours of the rocks are assigned higher weights. The final output feature map F_{output} is obtained by performing element-wise multiplication between the weighted map and the original image on a pixel-by-pixel basis:

$$F_{\text{output}} = \text{Sigmoid}(F_o \otimes F_{\text{input}}) \quad (4)$$

The CEM is the core component of the MLCEM module. As shown in Fig. 3, the main idea of CEM is to apply the sobel operator as described in Eq. (5), G_x and G_y to perform edge enhancement on the input

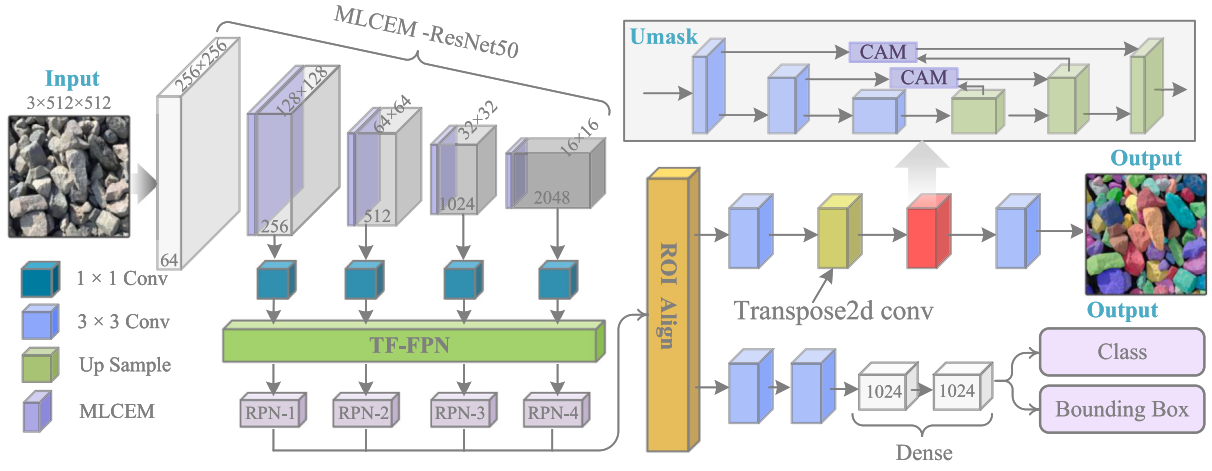


Fig. 1. CTURNet network architecture diagram.

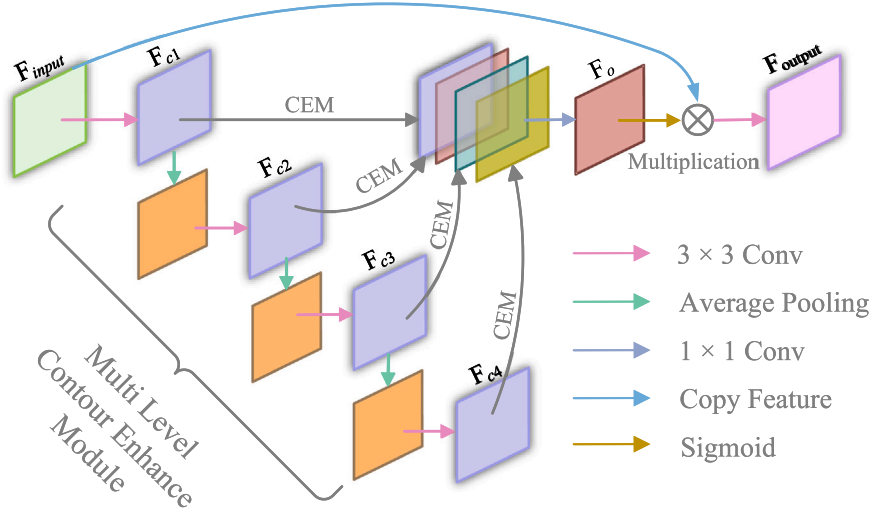


Fig. 2. Multi level contour enhance module.

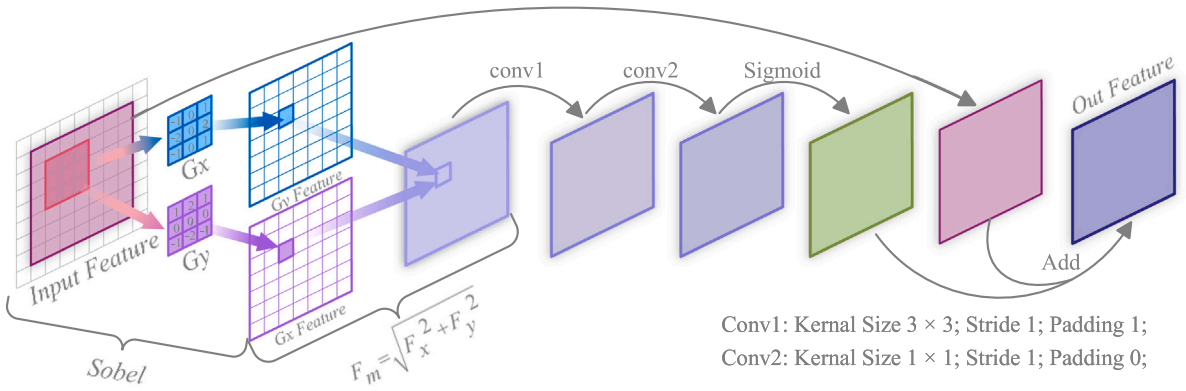


Fig. 3. Contour enhance module.

image, thereby obtaining the gradient information of image contours. The contour information in the x and y directions is then combined using the sum of squares in Eq. (6), resulting in a contour-enhanced feature map.

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} \quad (5)$$

$$F_m = \sqrt{F_{G_x}^2 + F_{G_y}^2} \quad (6)$$

where F_{G_x}, F_{G_y} represent the feature maps in the x and y directions, respectively, while F_m is the squared sum of the feature maps in the x and y directions.

Subsequently, 3×3 and 1×1 convolutional kernels are used for preliminary feature extraction and feature fusions, respectively. After

the convolution operations, a sigmoid activation function is introduced to enhance nonlinear features. Finally, an addition operation is performed to combine the enhanced contour features with the original features. This fusion approach effectively preserves the detail information of the original image and enables the model to better capture complex edge structures in the subsequent mask generation stage.

3.3. Transformer-based feature pyramid network

Feature Pyramid Networks (FPN) are an effective multi-scale feature fusion approach [47], capable of handling both local details and global semantic information in images. However, traditional FPN often suffer from limited channel capacity, which may result in the loss of small-scale rock features. Additionally, Due to the similar colors of rocks, traditional FPNs using element by element addition for feature fusion can easily lead to blurry details, thereby affecting detection accuracy. To address this issue, this paper introduces a Transformer branch into the FPN architecture. Inspired by the properties of Transformers and the structure of FPN [48], we propose a TF-FPN network specifically designed for multi-scale feature extraction of rock targets. By incorporating Transformer into the FPN framework, the proposed method establishes stronger global correlations across different scales, thereby significantly enhancing performance in complex object segmentation tasks. Specifically, in the encoding stage, the Transformer models the features through a global self-attention mechanism, enabling comprehensive capture of the multi-scale relationships among rock fragments in the image.

The TF-FPN module mainly consists of three parts: the feature sampling part, the Transformer-based feature extraction and interaction part, and the feature map reconstruction part. The core idea of the feature sampling module is to project features of different scales into a unified serialized representation space, aligning spatial information across scales within the same space. Then, through the Transformer's attention mechanism, cross-scale information interaction and semantic fusion are achieved, resulting in a unified serialized feature representation. To this end, we designed sampling windows of different sizes $k_i \in \{8 \times 8, 4 \times 4, 2 \times 2, 1 \times 1\}$ for feature maps extracted from the i th layer of the backbone (ResNet50), denoted as $F_i \in \mathbb{R}^{C \times W_i \times H_i}$, where $C=256$ and $(H_1, W_1) = (128, 128), (H_2, W_2) = (64, 64), (H_3, W_3) = (32, 32), (H_4, W_4) = (16, 16)$ as shown in Fig. 4. Specifically, larger sampling windows (e.g., 8×8) are used to aggregate global semantic information from low-level features, while smaller sampling windows (e.g., 1×1) compress detailed information from high-level features. This operation can be regarded as a sliding aggregation process similar to a sampling window. After sampling, each feature map is divided into $S_1 \in \mathbb{R}^{C \times 8 \times 8}, S_2 \in \mathbb{R}^{C \times 4 \times 4}, S_3 \in \mathbb{R}^{C \times 2 \times 2}$, and $S_4 \in \mathbb{R}^{C \times 1 \times 1}$. After flattening $S_i (i = 1 - 4)$, a total of $N = 1 + 2^2 + 4^2 + 8^2 = 85$ sampling blocks are obtained. The flattened sampling results from all scales are concatenated in hierarchical order to form the unified multi-scale feature sequence representation F_m .

$$F_m = \text{Concat}(A_{1,1}(F^4), \Gamma(A_{2,2}(F^3)), \Gamma(A_{4,4}(F^2)), \Gamma(A_{8,8}(F^1))) \quad (7)$$

where $A_{k,k}$ denotes the sliding window operation applied to the feature map with a window size of k and stride k for information extraction, Γ represents the flattening operation applied to the values extracted from the sliding windows, F^i denotes the feature map at the i th level, and F_m represents the fused feature obtained by aligning and concatenating the pixel values from all feature maps.

After obtaining F_m , it is from $F_m \in \mathbb{R}^{256 \times 85 \times 256}$ transformed to $F_m \in \mathbb{R}^{256 \times 256 \times 85}$, representing a sequence containing 256×256 tokens, each with a feature dimension of 85. The goal is to align spatial information from different scales within a unified representation space, allowing the model to perform global self-attention computation along the token dimension. This design overcomes the local receptive field limitation of convolutional structures, theoretically enabling the

capture of global dependencies and enhancing cross-scale consistency in the feature pyramid.

The second part is Transformer-based feature extraction and interaction, as shown in Fig. 5. Directly feeding excessively large feature vectors into the Transformer would reduce computational efficiency. Since our objective is to extract relationships among the four feature maps, the input features are first downsampled to obtain $F_d \in \mathbb{R}^{128 \times 128 \times 85}$. The downsampled feature map is then reshaped into a Transformer-compatible $F_d \in \mathbb{R}^{16384 \times 85}$, followed by an embedding process that adjusts its dimension to $F_e \in \mathbb{R}^{16384 \times 256}$. Before inputting the embedded features into the Transformer Encoder, positional encoding is applied using sine and cosine functions, producing F_p . Finally, F_p is fed into the Transformer Encoder to compute inter-feature relationships. Specifically, the input sequence F_p is linearly projected to generate query, key, and value matrices:

$$Q = F_p W_Q, \quad K = F_p W_K, \quad V = F_p W_V \quad (8)$$

where W_Q, W_K, W_V are learnable parameter matrices. The self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (9)$$

The third part is the feature map reconstruction stage. In this stage, the Transformer-encoded output sequence is upsampled to $256 \times 256 \times 85$, then reshaped and rearranged (inverse flattening) to restore it to the original feature map size before fusion. The reconstructed feature map is concatenated with the corresponding input feature map along the channel dimension, followed by a 1×1 convolution for dimensional adjustment. The output is then processed following the FPN [47] structure to produce the final multi-scale feature maps $P_1 - P_4$.

Through this design, TF-FPN effectively fuses multi-scale features and leverages the Transformer's global self-attention mechanism to enhance global dependency modeling among features, thereby achieving superior performance in complex object segmentation tasks.

3.4. Umask branch prediction

In image instance segmentation tasks, improving mask quality has long been a key research focus. In most current instance segmentation algorithms, the mask branch typically adopts a Fully Convolutional Network (FCN) architecture. However, traditional FCN structures exhibit several notable limitations: on one hand, they are generally shallow with limited feature refinement capability, making it difficult to fully exploit the rich deep semantic information within the ROI regions, thereby restricting the fine-grained representation of mask predictions. On the other hand, traditional FCN lack explicit feature enhancement mechanisms treating all channels equally failing to dynamically adjust feature responses based on the importance of different regions. This is particularly problematic in rock images where boundaries are blurred, textures are complex, and contrast is low, often resulting in poor segmentation performance for small or indistinct targets.

To address these issues, this paper proposes a novel mask branch structure named Umask. Building on the traditional FCN, Umask incorporates a U-Net architecture that performs progressive downsampling and upsampling to achieve multi-scale feature aggregation. Additionally, it integrates a Channel Attention Mechanism (CAM) [49] to dynamically model the importance of feature channels, significantly enhancing both spatial detail modeling and global semantic understanding. Through these design choices, Umask effectively mitigates the challenges faced by traditional FCN in rock instance segmentation, such as unclear boundary separation and missed small targets, thereby improving mask quality and instance segmentation accuracy.

Fig. 6 shows the architecture of the Umask network. Umask first takes the feature $F_{a1} \in \mathbb{R}^{256 \times 28 \times 28}$, extracted by ROI Align, as input. On

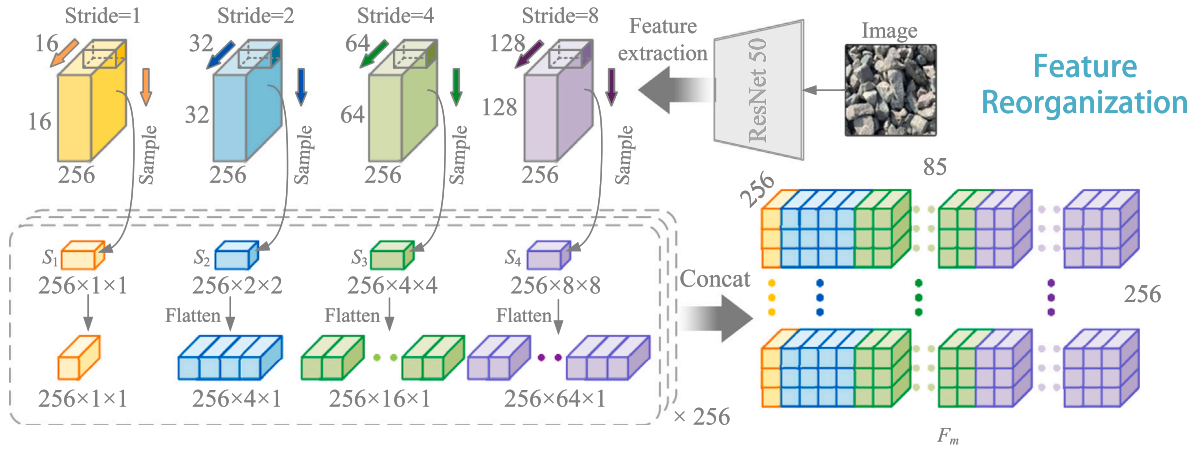


Fig. 4. Feature extraction and reconstruction process.

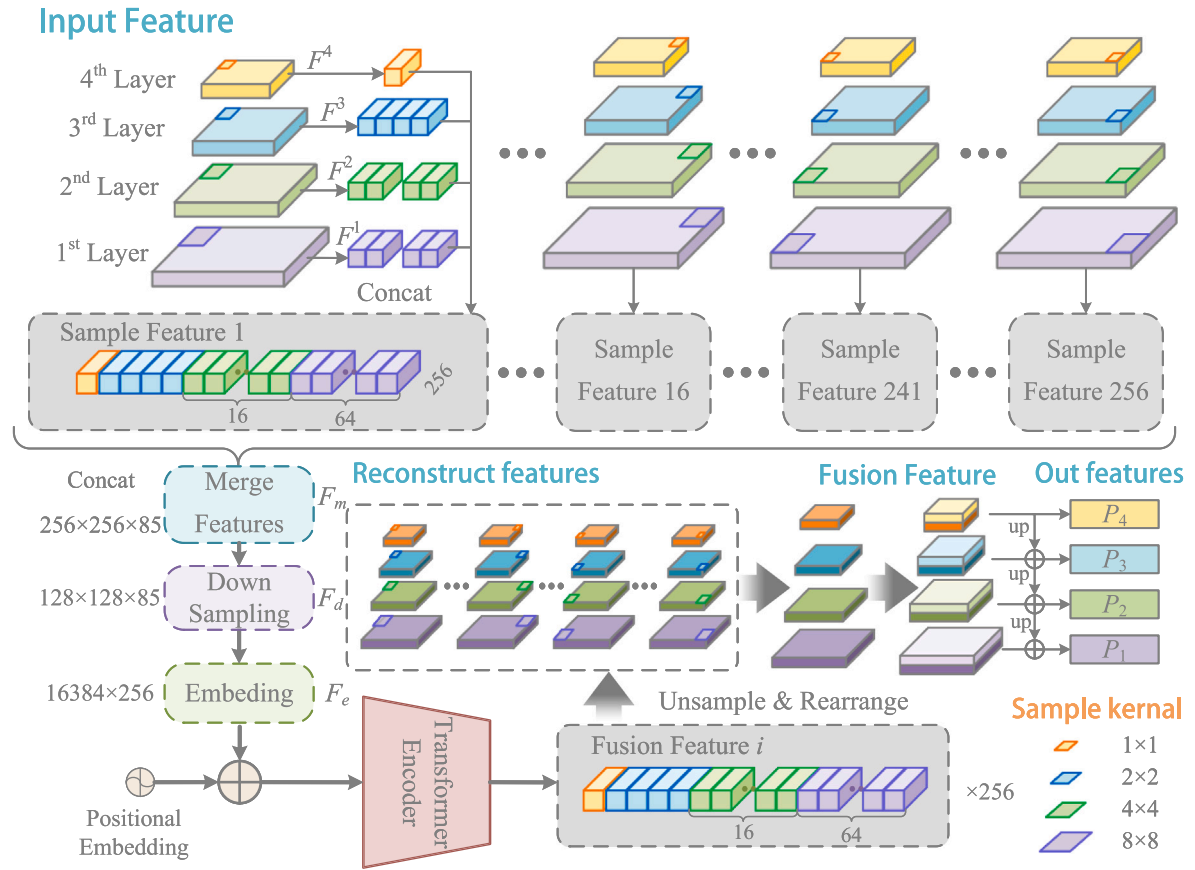


Fig. 5. Transformer-based feature pyramid network.

one hand, a skip connection is used to directly preserve the original features; on the other hand, the input enters an encoder path for feature downsampling and compression. After the first downsampling, the feature size is reduced from 28×28 to 14×14 while the channel number remains at 256, producing feature $F_{d1} \in \mathbb{R}^{256 \times 14 \times 14}$. Then, F_{d1} passes through two sets of 3×3 convolutions, LayerNorm, and ReLU activation to extract higher level semantic features, resulting in output $F_{a2} \in \mathbb{R}^{512 \times 14 \times 14}$. At this stage, the spatial resolution remains 14×14 while the channel number is increased to 512 to enhance feature representation capability.

Building on this, F_{a2} is passed through a second downsampling branch, further reducing the spatial size to 7×7 while maintaining 512 channels, resulting in feature F_{d2} . Through a series of convolution

operations, the channel number is first increased to 1024 and then reduced back to 512 to produce feature $F_{e2} \in \mathbb{R}^{512 \times 7 \times 7}$, capturing richer and more abstract contextual information. To restore spatial resolution, F_{e2} is upsampled back to 14×14 with the channel size maintained at 512. It is then fused with F_{a2} and processed through a CAM, producing the enhanced feature $F_{m2} \in \mathbb{R}^{1024 \times 14 \times 14}$. After two convolution layers, the feature F_{e1} is obtained, which is then fused with the original skip-connected feature F_{a1} and passed through another CAM module to produce $F_{m1} \in \mathbb{R}^{512 \times 28 \times 28}$. Finally, the fused feature F_{m1} is fed into a convolutional layer to generate $F_{mask} \in \mathbb{R}^{C \times 28 \times 28}$, achieving instance-level mask classification for each pixel. In the rock instance segmentation task, $C = 2$, so $F_{mask} \in \mathbb{R}^{2 \times 28 \times 28}$.

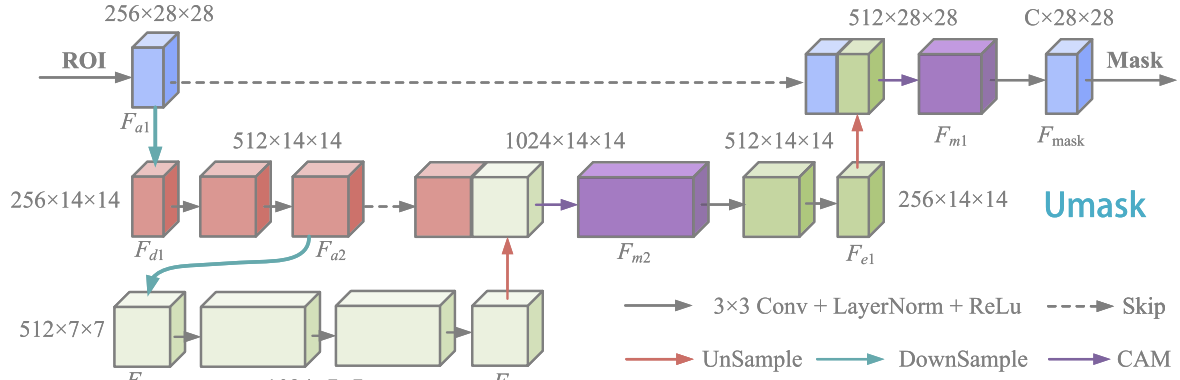


Fig. 6. Umask branch prediction module.

3.5. Loss functions

The loss function is used to measure the discrepancy between the model's predicted values and the ground truth. By minimizing the loss function, the model can optimize its parameters to improve prediction accuracy. In this paper, the loss function of CTURNet consists of three components: classification loss $\mathcal{L}_{cls}$, bounding box regression loss $\mathcal{L}_{box}$, and mask loss $\mathcal{L}_{mask}$.

$$\begin{cases} \mathcal{L}_{cls} = -\sum_{i=1}^N y_i \log(p_i) \\ \mathcal{L}_{box} = \sum_{i=1}^N \text{SmoothL1}(t_i - p_i) \\ \mathcal{L}_{mask} = -\sum_{i=1}^N y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \end{cases} \quad (10)$$

The total loss function is the weighted sum of the individual loss functions and is typically represented as:

$$\mathcal{L}_{total} = \mathcal{L}_{cls} + \lambda_{box} \mathcal{L}_{box} + \lambda_{mask} \mathcal{L}_{mask} \quad (11)$$

where $\mathcal{L}_{total}$ represents the total loss, while λ_{box} and λ_{mask} are weighting coefficients used to balance the contributions of different loss components to the total loss. In this study, λ_{box} is set to 0.5 and λ_{mask} is set to 1.

4. Experiments and results

4.1. Dataset acquisition

The rock image dataset used in this study was collected from a certain rock field in Anshan, Liaoning Province, China, with granite as the iron rock type. The images were captured using a smartphone equipped with a 12MP main camera. The original images possess high spatial resolution, to facilitate segmentation model prediction, preprocessing steps were applied as illustrated in Fig. 7: first, to avoid significant measurement errors caused by distortion at the image edges, the original images were center-cropped to a resolution of 3072×2048 pixels. Subsequently, to further optimize model performance, each high-resolution image was divided into six smaller images of 1024×1024 pixels. Considering GPU memory limitations and the need to accelerate training, the rock images were then downsampled to 512×512 pixels. These preprocessing steps help improve the training efficiency of the model.

A total of 150 high-resolution images were captured for this study, and manual polygon annotations of the rocks were performed using Labelme software. Due to the presence of very small or heavily occluded rock fragments, only clearly visible rocks were annotated. As a result, the subsequent segmentation algorithms may overlook particularly small or indistinct fragments.

After annotation, the JSON files generated by Labelme were converted into the VOC2012 dataset format. Following data selection, approximately 800 annotated rock images were obtained. To expand the dataset and enhance the model's generalization capability, data

augmentation techniques such as random flipping, random cropping, and random rotation were applied, increasing the dataset size to 1600 images. Additionally, to further improve the model's robustness, we also selected a publicly available rock instance segmentation dataset. This dataset [50] was collected from open-pit mining scenes and mainly contains bauxite images, with a total of 900 samples, which were used together with our dataset for model training. Ultimately, after shuffling, a total of 2500 rock images were used, with the dataset split into training, validation, and testing sets at a ratio of 7 : 1 : 2.

4.2. Evaluation metrics

In the rock particle size segmentation task, we evaluate the model's mask prediction performance using several metrics: Precision, Recall, Pixel Accuracy (PA), Intersection over Union (IoU), Mean Average Precision (mAP) and Mean Average Recall (mAR) as defined in Eqs. (12), (13), (14), (15), (16), (17) to evaluate the model's mask prediction performance. Precision measures the proportion of true positive predictions among all samples predicted as positive by the model; recall evaluates the proportion of actual positive samples correctly predicted as positive. PA represents the proportion of correctly classified pixels at the pixel level. IoU is the ratio of the intersection to the union between the predicted region and the ground truth region. AP refers to the average area under the precision-recall curve (Area Under Curve, AUC), reflecting the model's precision performance across different recall levels; AR represents the average area under the recall-precision curve, indicating the model's recall performance across different IoU thresholds. It should be noted that mAP is the average of AP, and mAR is the average of AR. Since this study focuses on single-class rock detection, mAP is equal to AP, and mAR is equal to AR.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

$$PA = \frac{TP + TN}{TP + TN + FP + FN} \quad (14)$$

$$IOU = \frac{\text{Area}(\text{Region}_p \cap \text{Region}_{gt})}{\text{Area}(\text{Region}_p \cup \text{Region}_{gt})} \quad (15)$$

$$AP = \sum_{i=1}^{n-1} (r_{i+1} - r_i) P_{interp}(r_{i+1}) \quad (16)$$

$$AR = 2 \int_{0.5}^1 \text{recall}(o) do \quad (17)$$

where TP (True Positive) refers to samples correctly predicted as positive; TN (True Negative) refers to samples correctly predicted as negative. FP (False Positive) indicates negative samples incorrectly

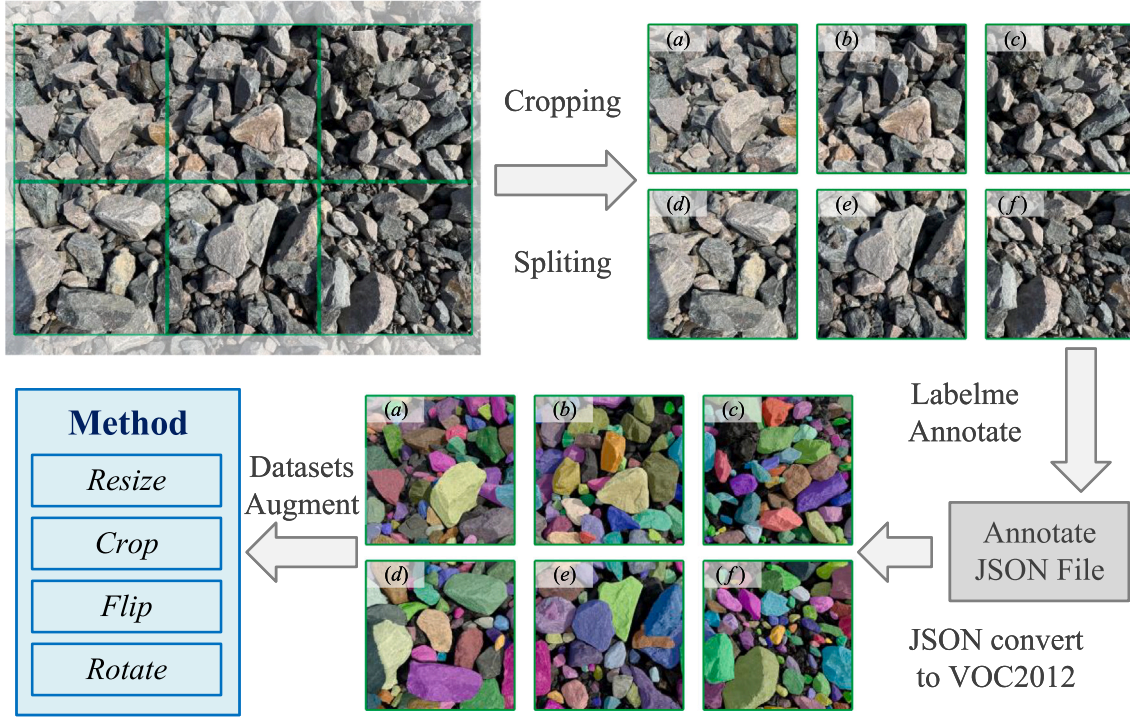
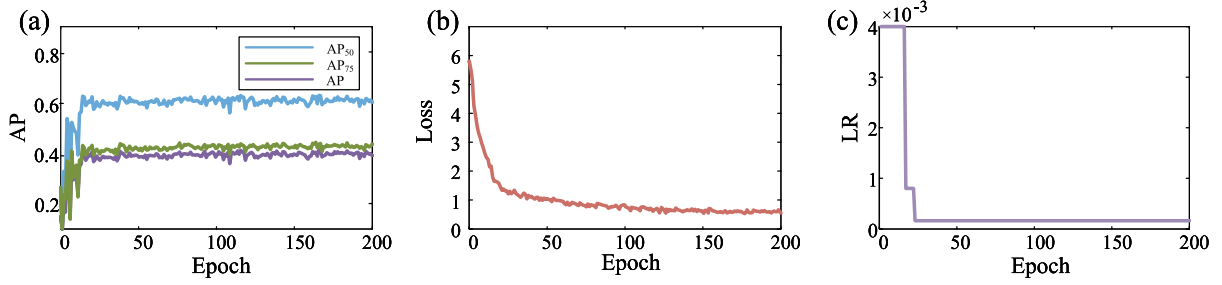


Fig. 7. Dataset acquisition process.

Fig. 8. Overall indicators during training: (a) Model predicted AP_{50} , (b) Loss value, (c) Learning rate.

predicted as positive, while FN (False Negative) refers to positive samples incorrectly predicted as negative. $Region_p$ denotes the predicted bounding box or mask region, and $Region_{gt}$ denotes the ground truth bounding box or mask region. In this study, AR is computed with parameters $Area = all$ and $maxDets = 100$, and AP is calculated based on an IoU threshold of 0.5 and 0.75.

4.3. Training results

The proposed CTURNet was implemented using the Python programming language and the PyTorch deep learning framework. Model training and evaluation were conducted on a computing platform equipped with the Ubuntu 20.04 LTS operating system, an Intel Xeon(R) Platinum 8171M CPU, and an NVIDIA Tesla A100 GPU. The SGD optimizer was selected as the gradient descent algorithm. A multi-step learning rate decay strategy was applied based on experimental experience and hardware capabilities, with the learning rate curve shown in Fig. 8(c). During training, the batch size was set to 8, and the number of epochs was set to 200. Each image was uniformly resized to 512×512 pixels.

The AP_{50} curve of the CTURNet network is shown in Fig. 8(a). As training progresses, the AP_{50} value rises sharply around the 25th epoch and stabilizes at approximately 0.62. This trend indicates that the model effectively learns feature representations in the early stages

of training and demonstrates good convergence. Fig. 8(b) presents the total loss curve of the network. At the beginning of training, the loss value is approximately 5.8, and it decreases rapidly within the first 50 epochs, eventually stabilizing around 0.7.

4.4. Model comparison and discussion

4.4.1. Comparison of test results

To evaluate the performance of the proposed CTURNet network, we first conducted experiments on a mixed dataset, followed by model stability tests on both a public dataset and a self-built dataset. The comparison models include Mask R-CNN [40], Cascade R-CNN [51], Hybrid Task Cascade (HTC) [52], MS R-CNN [53], and Mask2Former [43]. To ensure fairness, all models used the same backbone (ResNet-50), identical training strategies, and hyperparameter settings, and were trained and evaluated independently on each dataset.

First, a quantitative analysis was conducted, in the mixed dataset experiment, average performance metrics were computed for every 100 images out of 500 test samples, and the overall mean and standard deviation were then calculated. The results are shown in Table 1, while Tables 2 and 3 present the quantitative comparison results on the public and self-built datasets, respectively. To visually illustrate segmentation performance, five representative test samples were selected for qualitative visualization in Fig. 9, using model parameters trained on the

Table 1
Comparison of segmentation performance among different algorithms.

Model	AP	AP ₅₀	AP ₇₅	AR	IOU	PA	Recall	Precision
Mask R-CNN	37.25 ± 0.63	57.79 ± 0.51	40.10 ± 0.59	62.75 ± 0.65	83.72 ± 0.53	80.38 ± 0.81	82.62 ± 0.85	87.18 ± 0.71
Cascade R-CNN	38.48 ± 0.52	58.28 ± 0.71	41.39 ± 0.45	63.44 ± 0.75	85.67 ± 0.47	82.73 ± 0.62	83.75 ± 0.68	90.23 ± 0.72
HTC	39.28 ± 0.56	59.04 ± 0.68	42.03 ± 0.57	64.16 ± 0.64	86.17 ± 0.37	84.32 ± 0.45	84.03 ± 0.52	91.28 ± 0.49
MS R-CNN	38.80 ± 0.40	58.50 ± 0.74	41.82 ± 0.44	64.68 ± 0.52	85.93 ± 0.52	82.70 ± 0.64	83.35 ± 0.66	90.19 ± 0.86
Mask2Former	37.52 ± 0.74	58.20 ± 0.65	40.71 ± 0.65	62.59 ± 0.77	83.93 ± 0.58	81.70 ± 0.75	83.35 ± 0.78	87.69 ± 0.85
Ours	40.54 ± 0.54 (3.29 ↑)	61.82 ± 0.73 (4.03 ↑)	43.51 ± 0.62 (3.41 ↑)	65.12 ± 0.54 (2.37 ↑)	88.06 ± 0.32 (4.34 ↑)	85.15 ± 0.42 (4.77 ↑)	86.85 ± 0.68 (4.23 ↑)	93.23 ± 0.74 (6.05 ↑)

Table 2
Performance metrics on the public dataset.

Model	AP	AP ₅₀	AP ₇₅	AR	IOU	PA	Recall	Precision
Mask R-CNN	36.44	57.52	39.96	62.54	83.46	79.79	82.33	86.78
Cascade R-CNN	37.28	57.89	40.42	63.04	84.58	82.26	84.53	89.51
HTC	38.82	58.56	41.85	63.01	85.05	83.25	83.10	91.28
MS R-CNN	37.53	57.25	40.35	63.51	84.26	81.35	82.04	89.69
Mask2Former	36.51	58.05	39.00	62.58	83.45	80.25	83.29	86.91
Ours	39.63(3.19 ↑)	60.53(3.01 ↑)	42.78(2.83 ↑)	64.26(1.72 ↑)	86.41(2.95 ↑)	84.05(4.26 ↑)	85.26(2.93 ↑)	91.71(4.93 ↑)

Table 3
Performance metrics on the self-built dataset.

Model	AP	AP ₅₀	AP ₇₅	AR	IOU	PA	Recall	Precision
Mask R-CNN	38.36	58.93	41.25	64.12	84.96	81.62	83.75	88.54
Cascade R-CNN	39.46	58.35	42.72	64.62	86.86	83.55	84.84	91.51
HTC	40.25	61.17	43.46	65.24	87.22	84.57	85.89	92.59
MS R-CNN	39.88	59.64	42.07	64.74	87.11	83.86	84.19	91.24
Mask2Former	39.63	59.31	41.85	65.17	86.62	82.64	84.34	89.84
Ours	41.67(3.31 ↑)	63.01(4.08 ↑)	44.76(3.51 ↑)	66.37(2.25 ↑)	89.26(4.3 ↑)	86.51(4.89 ↑)	88.02(4.27 ↑)	94.38(5.84 ↑)

mixed dataset the first two images are from the public dataset and the last three are from the self-built dataset.

From the results in Tables 1, 2, 3, CTURNet achieves the best segmentation performance across the mixed, public, and self-built datasets. On the mixed dataset, CTURNet attains an AP of approximately 40.54%, about 3.29% higher than the baseline Mask R-CNN, demonstrating greater stability in boundary detection and segmentation accuracy. Cascade R-CNN and MS R-CNN achieve comparable overall performance, while HTC shows slight improvements in AP and AR but still performs below CTURNet. Although Mask2Former incorporates a Transformer architecture, it fails to fully exploit its global modeling advantages on rock images.

On the public dataset, the performance gap between models narrows, yet CTURNet continues to outperform others in key metrics such as AP and Precision, showing strong generalization ability. On the self-built dataset, CTURNet achieves an AP of approximately 41.67%, with Recall, PA, and Precision improving by about 4%–6% over the baseline, showing the most significant performance gains. The overall accuracy on the self-built dataset is higher than that on the public dataset, likely due to better image quality and labeling consistency.

Overall, CTURNet demonstrates superior segmentation accuracy and stability across all three data sources, validating the effectiveness and engineering potential of the proposed method for complex rock segmentation tasks.

Next, a qualitative analysis was performed. In Fig. 9, from top to bottom are the original rock image, Ground Truth, and the detection results of Mask R-CNN, Cascade R-CNN, HTC, MS R-CNN, Mask2Former, and CTURNet. It can be observed that, CTURNet outperforms other mainstream instance segmentation models in accurately detecting rock counts and boundary contours. As shown in Fig. 9(3) bottom right and (5) bottom left, CTURNet maintains good segmentation performance even under severe adhesion and blurred boundaries. HTC, Cascade R-CNN, and CTURNet achieve better results on small rock fragments; however, the first two still miss some rocks with unclear edges, as indicated by the white boxes in Fig. 9. Although Mask2Former employs a Transformer architecture to enhance global feature modeling, its

Table 4
Comparison of computational complexity and parameters of different segmentation algorithms.

Model	Backbone	Params (M)	GFLOPS (G)	FPS
Mask R-CNN	ResNet50	44	207	10.3
Cascade R-CNN	ResNet50	65	275	8.4
HTC	ResNet50	80	441	5.0
MS R-CNN	ResNet50	39	284	8.3
Mask2Former	ResNet50	49	226	9.7
Ours	ResNet50	85	465	4.8

segmentation of rock images tends to be incomplete due to complex textures and boundary details relying on local features (see Fig. 9(3)). Overall, CTURNet surpasses existing mainstream instance segmentation models, demonstrating its potential and advantages in practical applications.

4.4.2. Computational complexity analysis

The computational complexity of the proposed CTURNet was analyzed in terms of the number of parameters (Params) and floating-point operations (GFLOPs). The results show that the baseline model, Mask R-CNN, has approximately 44M parameters and 207 GFLOPs. On this basis, the three proposed key modules MLCM, TF-FPN, and UMask introduce 21M, 7M, and 13M additional parameters, and 124G, 46G, and 88G additional FLOPs, respectively. Consequently, the total parameter count of CTURNet reaches 85M, with a total computational cost of 465G. Although the overall complexity increases compared with the baseline, the performance improvement is substantial. Among the modules, MLCM contributes the most to fine-grained boundary feature modeling, UMask enhances mask quality through high-resolution feature reconstruction, and TF-FPN achieves efficient multi-scale feature fusion with relatively low computational overhead.

Table 4 presents a comparison of CTURNet and several other models in terms of computational complexity and performance. CTURNet has 85M parameters and 465G GFLOPs, which are comparable to those of HTC, while other models have lower parameter counts and

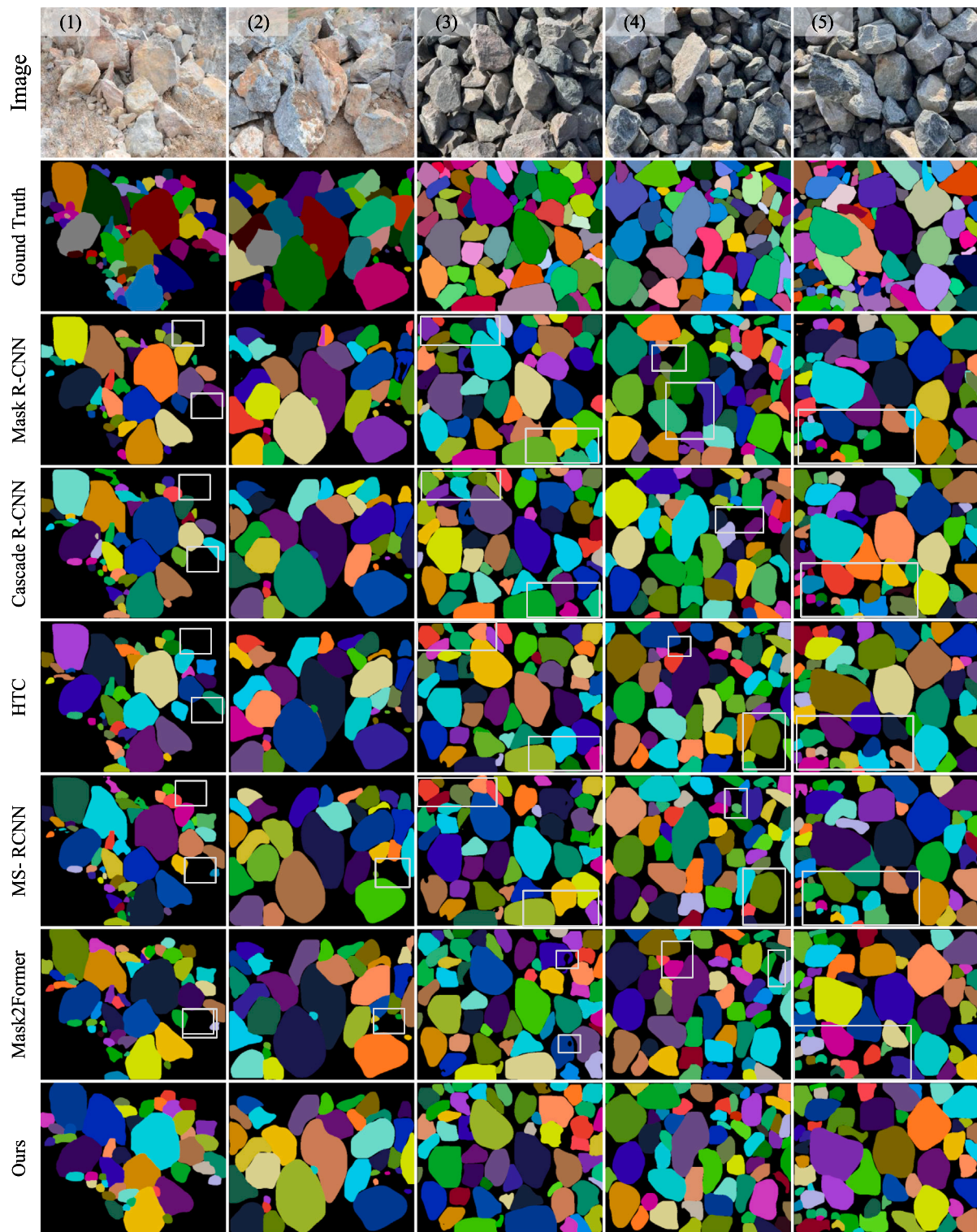


Fig. 9. Comparison of five rock image model results.

computational costs. Although both CTURNet and HTC exhibit higher computational complexity, they achieve significantly better detection accuracy than the other models. Moreover, under similar complexity, CTURNet still outperforms HTC in detection accuracy, demonstrating stronger feature extraction and representation capabilities. In summary, CTURNet achieves an excellent balance between computational complexity and detection accuracy, showing clear performance advantages and strong potential for practical applications.

4.4.3. Statistical analysis

To comprehensively evaluate the stability and reliability of the proposed CTURNet model and to verify whether its performance improvement over the SOTA model is statistically significant, multiple statistical analysis methods were employed, including 5-fold cross-validation, significance testing, paired t-test, and confidence interval analysis. First, 5-fold cross-validation was conducted to assess the

Table 5
Statistical analysis results.

Model	Fold	AP	IOU	Precision
HTC	1	38.83	85.92	90.96
	2	39.67	86.41	91.62
	3	39.41	86.12	91.27
	4	38.79	85.86	90.81
	5	39.74	86.55	91.72
	Mean±Std	39.28 ± 0.56	86.17 ± 0.37	91.28 ± 0.49
CTURNet	1	40.64	87.60	92.98
	2	39.92	87.51	92.61
	3	41.21	87.78	93.05
	4	40.63	87.45	94.19
	5	40.42	88.14	93.34
	Mean±Std	40.54 ± 0.54	87.69 ± 0.34	93.23 ± 0.74
Paired t-tests	<i>t</i>	4.47	8.32	6.12
	<i>p</i>	0.002	0.000033	0.00028

model's generalization ability under different data partitions. Specifically, the dataset was randomly divided into five subsets; in each iteration, one subset was used as the validation set while the remaining four were used for training. Table 5 presents the average performance results obtained after five repetitions. The HTC model was selected for comparison because it has a similar number of parameters to CTURNet and has demonstrated stable accuracy in previous detection tasks.

Based on the cross-validation results, significance tests were further conducted on key metrics, including AP, IoU, and Precision, with a significance level set to $p < 0.05$. The results show that CTURNet significantly outperforms HTC in AP, IoU, and Precision, with all differences being statistically significant. In particular, the *p*-values for IoU and Precision are extremely small (0.000033 and 0.00028, respectively), indicating that CTURNet achieves substantial improvements in these critical metrics. Further 95% confidence interval analysis reveals that CTURNet exhibits smaller performance fluctuations across all indicators compared to HTC, suggesting that CTURNet not only achieves higher mean performance but also demonstrates greater stability and consistency across multiple experiments.

4.5. Ablation study

4.5.1. Ablation study of modules

To verify the effectiveness of the CTURNet architecture, we conducted a series of ablation experiments to evaluate the contribution of each module to the overall model performance. The CTURNet network consists of three core modules: the Multi-Level Contour Enhancement Module (M), the TF-FPN Module (T), and the UMask Segmentation Module (U). In the experiments, eight model variants were constructed: the baseline model (Mask R-CNN), single-module models (Net1: Baseline+M, Net2: Baseline+T, Net3: Baseline+U), dual-module combinations (Net4: Baseline+M+T, Net5: Baseline+M+U, Net6: Baseline+T+U), and the complete CTURNet model. This setup allows analysis of both the independent effects and synergistic gains of each module. All models were trained on the same dataset with identical training strategies and hyperparameter settings to ensure fair comparison. Quantitative evaluations using AP, AR, IoU and other related metrics were performed to validate the effectiveness of each module in boundary recognition and multi-scale feature fusion. The detailed comparison results are presented in Table 6.

From the ablation results in Table 6, the overall performance of CTURNet improves consistently with the gradual introduction of each module, indicating that all components contribute positively to the network architecture. Compared with the baseline model (Mask R-CNN), the complete CTURNet achieves improvements of approximately 3.29% in AP, 2.37% in AR, 4.34% in IoU, 4.77% in PA, 4.23% in Recall, and 6.05% in Precision, demonstrating significant gains in segmentation accuracy and detection recall.

For single module models, adding MLCEM (Net1) increases AP from 37.45% to 38.21% and Precision to 90.13%, showing that the MLCEM effectively alleviates boundary ambiguity and improves fine-detail recognition. Incorporating TF-FPN (Net2) yields a slight improvement in AP₇₅, suggesting that the Transformer branch contributes to high-precision segmentation. Adding UMask (Net3) also enhances overall performance, with a more pronounced increase in IoU, indicating that the U-Net based mask branch refines object boundaries.

Among dual module combinations, Net4 (M+T) and Net5 (M+U) perform best. In particular, Net4 achieves an IoU of 87.31%, showing that the synergy between MLCEM and TF-FPN strengthens both boundary features and global semantic representations. Although Net6 (T+U) performs slightly lower in precision than Net4, it still surpasses single-module models, confirming the complementarity among modules. Ultimately, the full CTURNet achieves the best results across all metrics, demonstrating that the three modules form an effective multi-level collaborative fusion mechanism.

Overall, MLCEM enhances boundary clarity and segmentation precision, TF-FPN improves cross scale semantic modeling and feature consistency, and UMask refines fine-grained mask prediction. The combined use of these modules significantly improves the model's overall performance and stability, confirming the rationality of CTURNet's modular design and its strong synergistic benefits.

4.5.2. Selection of balance parameters in the segmentation loss function

To verify the rationality of the weight parameter settings in the segmentation loss, we conducted systematic ablation experiments on different weight combinations using the mixed dataset. Specifically, within the range of λ_{box} and $\lambda_{\text{mask}} \in [0.5, 1.5]$, the model's mask average precision (AP) was compared under various configurations.

The experimental results are shown in Table 7, the model achieves the best segmentation performance (AP = 40.42) when $\lambda_{\text{box}} = 0.5$ and $\lambda_{\text{mask}} = 1.0$. When λ_{box} increases, the bounding box regression loss dominates the overall optimization, causing the model to focus more on object localization and reducing the mask branch's ability to learn fine-grained boundary features. Conversely, when λ_{mask} is too large, the model overemphasizes mask optimization, leading to decreased localization accuracy. In summary, $\lambda_{\text{box}} = 0.5$ and $\lambda_{\text{mask}} = 1.0$ provide the optimal balance between detection and segmentation tasks. Therefore, this configuration is adopted as the final setting in all experiments.

4.6. Estimation of rock equivalent diameter

The rock fragments produced after blasting are generally assumed to be uniform and typically exhibit ellipsoidal shapes. In practical engineering scenarios, engineers usually estimate the size of rock fragments by averaging the lengths of their three principal axes. To apply rock segmentation results to the task of rock size measurement, it is necessary to calculate the equivalent diameter of each fragment based on the detection output, thereby obtaining the overall size distribution of the rocks. According to the studies by Kemeny [54] and Ye [55], a relationship has been established between the equivalent diameter d in the 2D plane and the parameters of the best-fitted outer ellipse.

First, CTURNet is used to extract the mask of each individual rock fragment. Then, OpenCV's algorithm library is employed to compute the area A and perimeter C of the mask. Using these values, the major axis a and minor axis b of the best-fitted ellipse in the 2D plane can be calculated using Eq. (18), and their geometric interpretation is illustrated in Fig. 10.

$$\begin{cases} a = \frac{1}{2} \cdot \left(\frac{C}{\pi} + \sqrt{\frac{C^2}{\pi^2} - \frac{4A}{\pi}} \right) \\ b = \frac{1}{2} \cdot \left(\frac{C}{\pi} - \sqrt{\frac{C^2}{\pi^2} - \frac{4A}{\pi}} \right) \end{cases} \quad (18)$$

Table 6
Evaluation results of ablation experiment.

Model	AP	AP ₅₀	AP ₇₅	AR	IOU	PA	Recall	Precision
Baseline	37.25	57.79	40.10	62.75	83.72	80.38	82.62	87.18
Net1	38.21	59.32	41.35	64.02	85.25	83.72	84.65	90.13
Net2	37.71	60.48	41.61	63.68	84.92	80.47	83.55	90.36
Net3	37.72	59.89	41.02	63.89	86.50	81.79	85.78	91.44
Net4	39.83	61.32	42.87	63.81	87.31	83.68	83.56	92.44
Net5	39.44	61.00	42.74	64.57	86.89	84.86	84.98	92.73
Net6	39.20	59.32	42.48	63.60	87.10	83.16	85.42	91.72
CTURNet	40.54(3.29 ↑)	61.82(4.03 ↑)	43.51(3.41 ↑)	65.12(2.37 ↑)	88.06(4.34 ↑)	85.15(4.77 ↑)	86.85(4.23 ↑)	93.23(6.05 ↑)

Table 7
Comparison of experimental results for balance parameter selection.

λ_{box}	λ_{mask}	AP
0.5	1.0	40.54
0.5	1.5	39.63
1.0	0.5	38.54
1.0	1.0	38.63
1.0	1.5	39.38
1.5	0.5	37.77
1.5	1.0	38.19

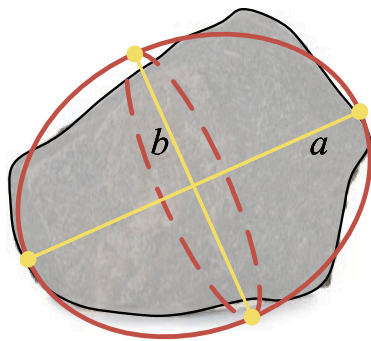


Fig. 10. The best-fitting outer ellipse of the particle mask.

Finally, the equivalent diameter d of the 2D shape is calculated using Eq. (19), and the actual equivalent radius of the physical object is then obtained by applying the image-to-real-world scaling factor ϵ .

$$\begin{cases} d = 1.16 \sqrt{1.35 \frac{a}{b}} \\ d_f = \epsilon d \end{cases} \quad (19)$$

Since a monocular camera was used for data collection in this paper, variations in shooting distance across different scenes made full geometric calibration difficult. Therefore, a reference-based proportional calibration method was adopted, in which a ruler of known length was placed in the scene. The pixel length L_{pixel} of the ruler was measured in the image and combined with its actual physical length L_{real} to calculate the pixel-to-real-world distance ratio:

$$\epsilon = \frac{L_{\text{real}}}{L_{\text{pixel}}} \quad (20)$$

Considering that the camera has nearly identical pixel density in the horizontal and vertical directions ($f_x \approx f_y$), only the x -direction scaling factor was used as a representative value. This method, which has been adopted in several related studies [31,55–57], enables reliable scale conversion without the need for complex camera calibration. In this paper, tape measures of varying lengths were used as reference scales at the blasting site, and original rock images were captured as shown in Fig. 11. It presents a comparison between CTURNet detection results and manual annotations. Six images were used as test samples, displayed from top to bottom as follows: the original rock image, the

manually annotated image, the rock segmentation result predicted by the CTURNet model, and the overlay of the predicted result on the original image. The tape measures in the images, from left to right, measure 0.5 m, 1 m, 0.5 m, 0.5 m, 0.5 m, and 1.3 m, respectively, serving as reference standards. The comparison clearly shows that the CTURNet model accurately predicts rock sizes in the images, with the shapes and positions of the predicted fragments closely matching those of the manually annotated rocks. This indicates that the CTURNet algorithm proposed in this study demonstrates high accuracy in rock detection tasks.

Based on the rock block regions detected by CTURNet, OpenCV was used to calculate the perimeter C and area A of each rock mask region. The rock equivalent size prediction model was then applied to estimate the rock size distribution curve. Through an automated processing algorithm, the equivalent sizes of all detected rocks were computed and compared with manually measured sizes to generate the cumulative rock size distribution curve shown in Fig. 12. The red and blue lines represent the cumulative size distribution curves of manual annotation and model prediction, respectively. To enhance the detail of the comparison, Table 8 lists the equivalent diameters and corresponding differences for six sample images at 10% cumulative increments, comparing the model's predictions with manual annotations.

As shown in Fig. 12 and Table 8, the particle size distribution of the six rock images is mainly concentrated between 30–350 mm, and the PSD curves predicted by the model are highly consistent with the manually labeled results. Overall, CTURNet exhibits small prediction errors and strong fitting capability for median particle size. For most samples, the predicted PSD curves closely overlap with the manual results in the D10–D90 range. (D_x represents the particle size corresponding to the cumulative percentage of $x\%$.)

From a local perspective, in the small particle size range (D10–D30), the predicted values are generally slightly lower than the manual annotations, with an error range of approximately 2–7 mm. This is mainly due to blurred boundaries or shadow occlusion in some fine fragments, causing incomplete mask recognition and underestimation of particle size. In the medium particle size range (D40–D60), the predicted results are closest to the manual curves, suggesting that the model maintains strong perceptual and semantic consistency at this scale. In the large particle size range (D70–D90), several samples (e.g., Samples 3, 5, and 6) show an overestimation trend, with errors around 10–15 mm. This mainly occurs when light-colored cracks or shadow interference between adjacent rocks cause over-segmentation during boundary detection, resulting in a slight rightward shift of the PSD curve in the large particle size region.

In addition, discrepancies increase gradually beyond D70, particularly in images containing more large rocks (e.g., Samples 1 and 3), where deviations near D50 are more pronounced. This is because large rocks are fewer in number, and individual errors have a greater influence on the overall distribution. Conversely, in images with smaller rock sizes and larger sample counts, the differences between model and manual results are smaller, demonstrating higher fitting accuracy. Except for Sample 1, all other images show errors within 10 mm in the D10–D70 range.

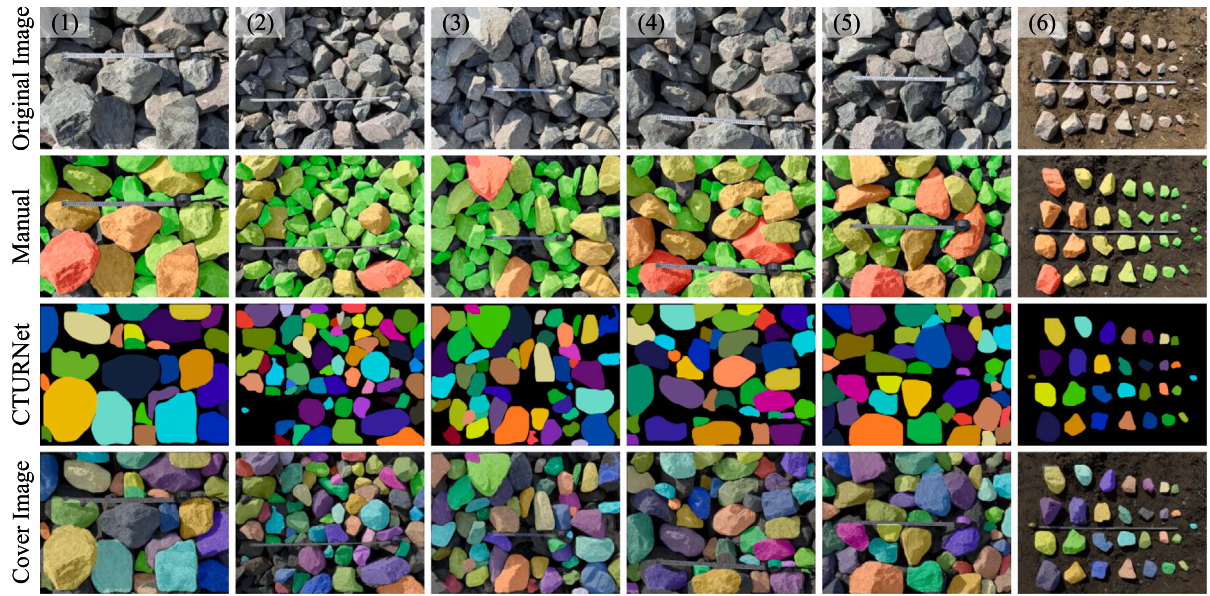


Fig. 11. Image comparison analysis (1–6 represent different on-site rock images).

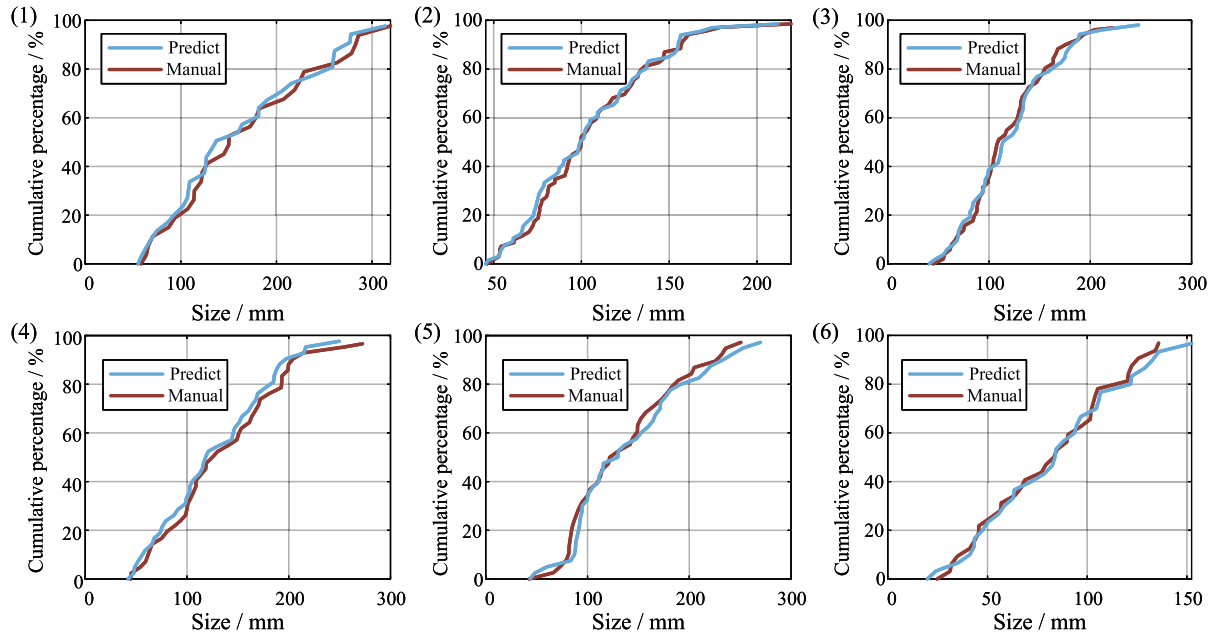


Fig. 12. Cumulative PSD curves of model prediction and manual segmentation.

To comprehensively evaluate model performance, the relative error index was introduced:

$$E_r = \frac{|M - P|}{M} \quad (21)$$

Where, E_r denotes the relative error, while M and P represent the manually measured and model-predicted equivalent radius sizes, respectively. After calculating and organizing the measurement data, we found that, across all cumulative percentages, the relative error between the predicted rock sizes and the manually measured actual equivalent diameters remains below 10%. This strongly demonstrates the practical applicability of the proposed network.

In summary, the main sources of PSD prediction error in CTURNet are small-fragment omissions and large-particle over-segmentation caused by blurred boundaries and uneven illumination. Despite minor local deviations, the overall error remains within a controllable range, and the PSD curves maintain high overlap near D50, indicating that

the proposed model accurately reflects rock particle size distribution characteristics and provides a reliable foundation for post-blast size statistics and fragmentation evaluation.

The size distribution of the rocks was further statistically analyzed in this study. We defined 15 size intervals, each with an interval width of I , and calculated the distribution using the following equation:

$$I = \frac{d_{\max} - d_{\min}}{n} \quad (22)$$

where $d_{\max}$ and $d_{\min}$ represent the maximum and minimum rock sizes in the image, respectively, and n is the number of intervals.

Using this method, we analyzed six collected images to determine the number of rock fragments within each size interval. Additionally, we calculated the mean and variance of rock sizes based on both manual annotations and CTURNet predictions, and plotted the Gaussian distribution curves of the rock size data.

Table 8
The size corresponding to the cumulative percentage (mm)

Image ID	Method	The size corresponding to the cumulative percentage (mm)									
		10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
1	Manual	63.8	96.1	115.6	125.9	145.2	178.6	217.8	251.2	286.4	319.4
	Predict	62.0	93.3	107.8	127.3	135.6	181.1	205.1	266.1	274.9	314.6
	Error	1.8	2.8	7.8	-1.4	9.6	-2.5	12.7	-14.9	11.5	4.8
2	Manual	61.7	75.2	80.9	91.7	98.9	108.8	124.5	134.2	157.3	220.6
	Predict	61.2	73.1	75.7	89.6	99.3	109.1	122.0	135.9	155.3	211.8
	Error	0.5	2.1	5.2	2.1	-0.4	-0.3	2.5	-1.7	2.0	8.8
3	Manual	67.2	85.7	94.9	103.3	102.9	128.6	136.2	155.4	171.3	234.1
	Predict	67.9	83.9	93.7	103.5	112.6	130.7	137.8	164.3	184.4	252.6
	Error	-0.7	1.8	1.2	-0.2	-9.7	-2.1	-1.6	-8.9	-13.1	-18.5
4	Manual	61.8	84.8	100.2	108.9	125.0	151.2	167.8	193.1	203.8	280.6
	Predict	57.6	78.1	98.6	105.9	118.3	146.3	162.8	184.8	197.6	263.8
	Error	4.2	6.7	1.6	3.0	6.7	4.9	5.0	8.3	6.2	16.8
5	Manual	80.8	84.3	92.9	109.9	121.9	148.9	154.9	185.3	226.0	254.6
	Predict	87.1	91.2	94.7	111.1	129.8	152.5	171.5	191.8	233.1	272.0
	Error	-6.3	-6.9	-1.8	-1.2	-7.9	-3.6	-16.6	-6.5	-7.1	-17.4
6	Manual	35.6	48.3	56.6	66.7	83.3	90.6	102.2	110.6	125.1	141.1
	Predict	36.3	52.5	59.2	70.2	83.8	93.8	104.6	121.7	132.9	152.4
	Error	-0.7	-4.2	-2.6	-3.5	-0.5	-3.2	-2.4	-11.1	-7.8	-11.3

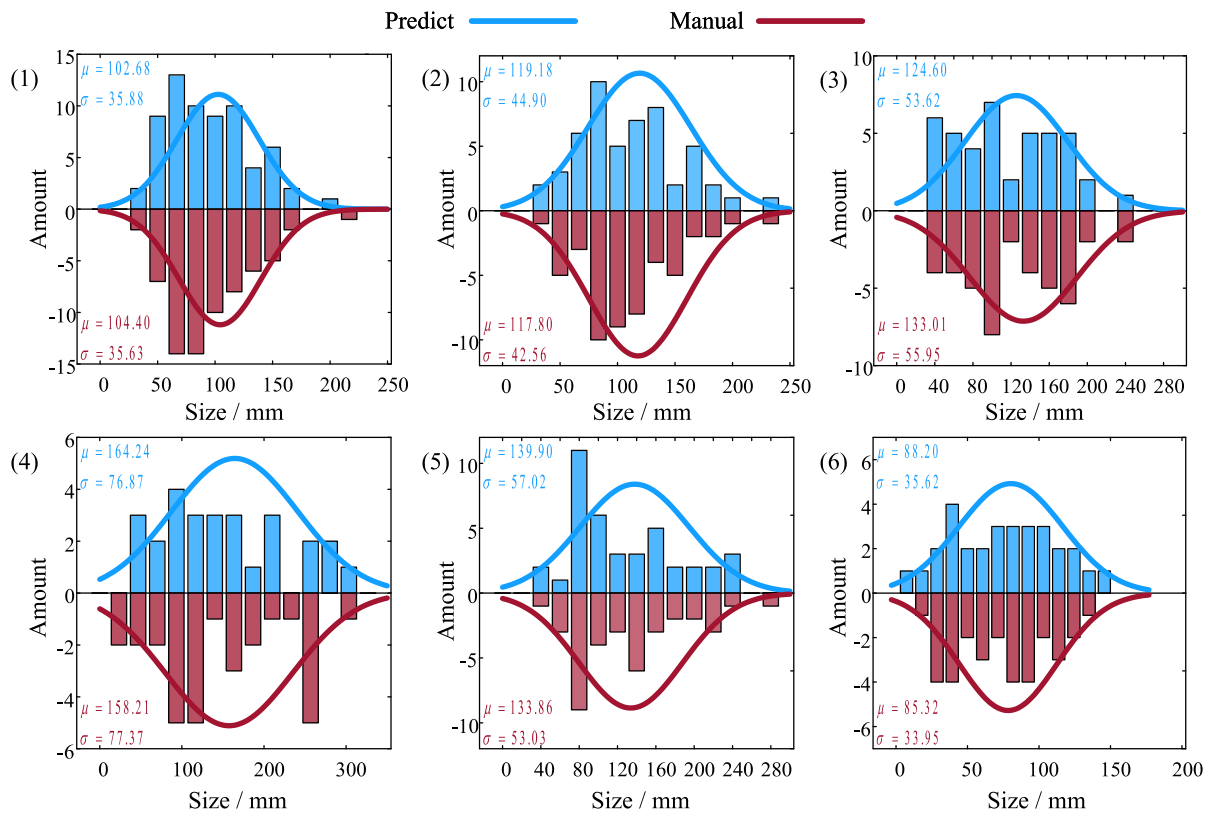


Fig. 13. PSD comparison between model prediction and manual segmentation.

In each subplot of Fig. 13, the blue bar graph represents the rock size distribution predicted by the CTURNet network, while the red bar graph corresponds to the manually annotated rock size distribution. As observed, the predicted size distribution closely aligns with the ground truth across all subplots. The average deviation of the distribution means is within 7 mm, and the variance difference is within 2 mm, further confirming the accuracy of CTURNet in rock size prediction.

4.7. Practical engineering application

The distribution of fragmentation plays a crucial role in evaluating blasting energy utilization efficiency, optimizing blasting parameters, and controlling the proportion of fine particles [58]. To verify the effectiveness of the predicted particle size distribution (PSD) in engineering

decision-making, a field case study was conducted at a granite open-pit mine in Xiuyan, Anshan, Liaoning Province (Fig. 14), with the target of obtaining rock aggregates with $D_{50} = 100$ mm. Under identical bench and geological conditions, post-blast rock pile images (Group 1) were collected. After scale calibration, the PSD was extracted using the proposed segmentation method, yielding $D_{50} \approx 140$ mm, which closely matched the on-site screening result of $D_{50} \approx 143$ mm. Therefore, the predicted PSD indicates that the rock pile is generally coarse, with a higher proportion of large fragments than the design target, suggesting that greater fragmentation energy is required to improve the size distribution structure.

According to literature and engineering experience, the main factors influencing PSD include the unit explosive consumption, the ratio of

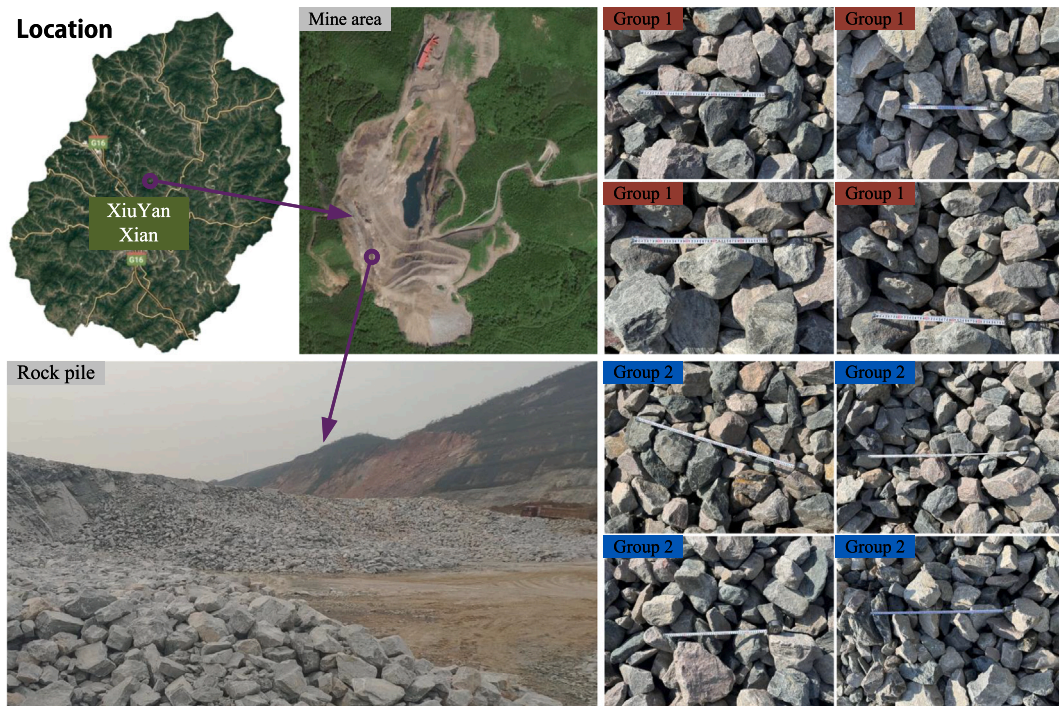


Fig. 14. XiuYan Mining area and post-blast rock size optimization analysis.

Table 9

Comparison of main blasting parameters.

Test group	Hole diameter (mm)	Spacing (m)	Hole depth (m)	Burden (m)	Unit explosive consumption (kg/m ³)
Group 1	160	7.5	16	5.0	0.36
Group 2	200	8.3	16	5.0	0.39

burden to spacing, the charge weight per hole or hole depth, and the charging structure. Based on the initial prediction results, targeted adjustments were made to these parameters, followed by a second set of comparative experiments (see Table 9): Group 1 ($\Phi=160$ mm, Spacing = 7.5 m, Burden = 5.0 m, $q = 0.36$ kg/m³) was adjusted to Group 2 ($\Phi=200$ mm, Spacing = 8.3 m, Burden = 5.0 m, $q = 0.39$ kg/m³), while the B/S ratio was reduced from 0.67 to 0.60. After adjustment, images from Group 2 were collected to recalculate the PSD. The results showed a $D_{50} \approx 80$ mm, consistent with the on-site screening data, demonstrating that the predicted PSD can directly guide and validate the optimization of blasting parameters.

In the rock transportation phase, D_{10} , D_{50} , and D_{90} were used as criteria for assessing shovelability and transport compatibility. The PSD from the initial blasting indicated excessive large fragments and insufficient uniformity, suggesting that small-bucket equipment would be unsuitable and that secondary crushing should be reserved. After parameter optimization, the PSD showed an increased proportion of small and medium-sized fragments, allowing the use of loading equipment with a higher fill factor while reducing the risk of secondary breaking and blockage. This improvement enhances loading efficiency and transport continuity. In summary, the case study demonstrates that the proposed algorithm not only achieves high segmentation accuracy but also enables a practical prediction parameter adjustment retesting closed-loop application in real mining environments, significantly reinforcing the engineering relevance of this research.

5. Discussion

5.1. Research and practical engineering significance of the proposed method

Rock instance segmentation and particle size distribution (PSD) estimation are fundamental tasks in blasting optimization and rock

fragmentation analysis. Traditional methods often suffer from boundary ambiguity, over-segmentation, and missed detections under complex backgrounds, illumination variations, and occlusions, limiting their accuracy. The proposed CTURNet integrates MLCem, TF-FPN, and the UMask module, significantly improving boundary perception, fine-detail recognition, and cross-scale semantic consistency. This provides a systematic approach for rock image segmentation in complex geological environments and offers theoretical contributions to the study of rock image segmentation and feature fusion.

From an practical engineering perspective, CTURNet provides valuable support for optimizing blasting operations and controlling aggregate size in open-pit mines. Accurate PSD prediction can effectively improve blasting efficiency and safety, optimize the ore extraction process, and reduce costs. Beyond the mining industry, CTURNet also has strong potential for application in other fields requiring precise object segmentation and feature extraction, such as construction and civil engineering.

5.2. Generalization discussion

To evaluate the generalization performance of the CTURNet model under different environmental conditions, four representative lighting scenarios were selected: sunny, cloudy, dusk, and rainy, as shown in Fig. 15.

The figure shows the original images, Ground Truth, and CTURNet segmentation outputs. The results indicate that CTURNet can consistently identify rock boundaries under varying illumination and surface moisture conditions, with segmentation results highly consistent with Ground Truth. In particular, under cloudy and rainy conditions, despite reduced surface brightness and increased reflection, the model can



Fig. 15. Segmentation results of rocks under different illumination conditions.

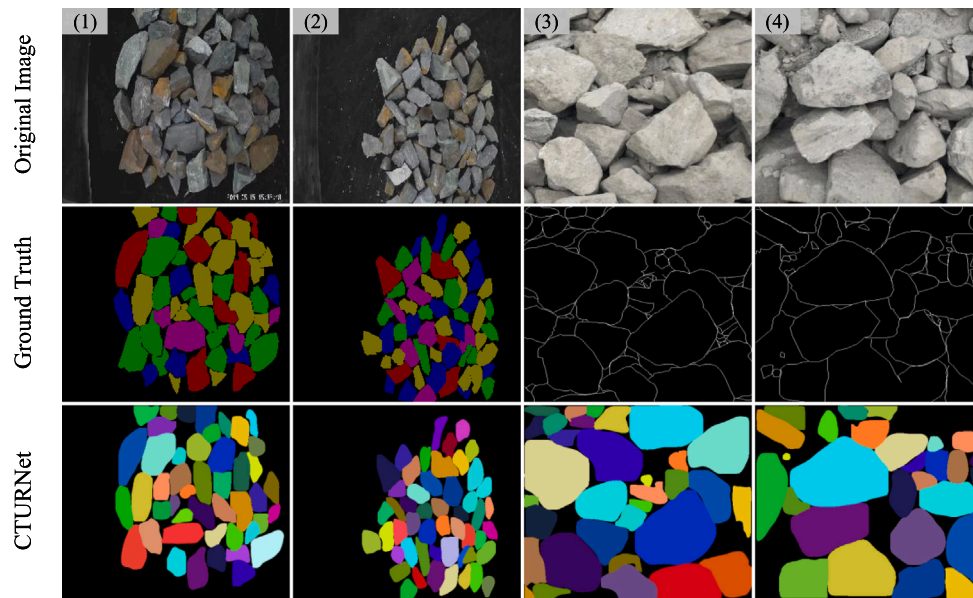


Fig. 16. Segmentation results of different rock types.

still accurately distinguish the boundaries of adjacent rocks, demonstrating that the proposed Multi-Level Contour Enhancement Module (MLCEM) maintains strong boundary sensitivity even in low-contrast environments.

Furthermore, to validate CTURNet's generalization ability across different lithological conditions, we tested the model on a multi-type rock image dataset collected from other studies. As shown in Fig. 16, panels (1) and (2) [59] present segmentation results on five representative rock types (limestone, hematite, ferric carbonate, chlorite, and ferrous rocks), while panels (3) and (4) [8] show test samples from a copper-molybdenum mine site. The results demonstrate that CTURNet can accurately segment individual rock blocks across various rock types, showing strong adaptability to variations in color, texture, and shape. The mask boundaries are clear and largely consistent with Ground Truth. Even in the copper-molybdenum mining scenario, the model maintains stable segmentation performance. These findings confirm that the proposed CTURNet exhibits strong generalization and

robustness across lithologies and data sources, providing experimental evidence for its potential application in different mining areas.

5.3. Failure modes and method limitations

Although CTURNet achieves strong overall performance in rock instance segmentation and PSD estimation, certain limitations remain under specific conditions, as summarized below:

(1) Weak recognition of small-sized rocks, as shown in the white boxes in Fig. 17. When rock particles are small or partially occluded, their features may be lost during multi-level downsampling, making it difficult for the model to effectively capture fine boundary information, which can lead to missed or incomplete detections.

(2) Over-segmentation in the presence of light cracks, shadows, reflections, or discontinuous textures on rock surfaces, as illustrated by the yellow boxes in Fig. 17.

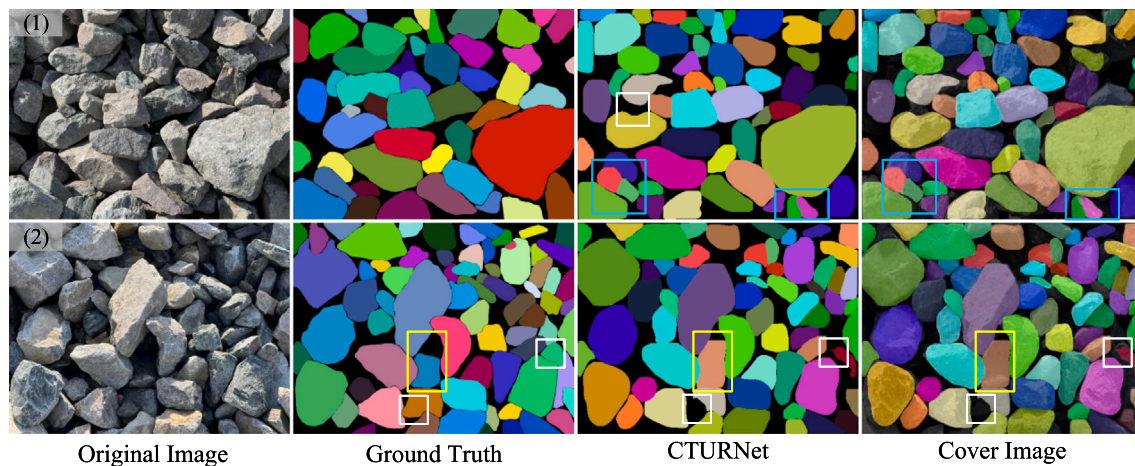


Fig. 17. Failure case examples.

(3) Under uneven illumination or when rocks have multiple angular facets, the boundary gradients on the rock surface become more pronounced, causing the model to misidentify different regions of the same rock as multiple independent instances, which leads to over-counting, as shown in the blue boxes in Fig. 17.

(4) Limited generalization, as the model's performance may vary across different environments or geological conditions due to the constraints of the training dataset.

Overall, these limitations mainly stem from visual ambiguity and texture complexity in the images, as well as the model's localized over-response in scale perception and boundary determination. However, their impact on the overall detection accuracy and results is relatively limited. In most cases, CTURNet still provides accurate segmentation and recognition performance.

5.4. Advantages and disadvantages of the model

The CTURNet model demonstrates significant advantages in rock instance segmentation and particle size distribution (PSD) estimation tasks. First, the model achieves outstanding segmentation accuracy. Through the MLCem and the UMask module, it effectively improves the segmentation precision of rocks with blurred edges, accurately capturing fine details even in challenging conditions such as complex backgrounds, occlusion, and uneven illumination. Second, benefiting from the TF-FPN, CTURNet excels in cross-scale information integration, overcoming the local receptive field limitations of traditional convolutional networks and enhancing semantic consistency and robustness. The model also exhibits excellent generalization performance, maintaining stable results across different mining sites and geological conditions, demonstrating strong adaptability. Moreover, CTURNet provides valuable engineering applicability by enabling accurate PSD prediction, which supports blasting process optimization and operational decision-making.

Despite its strong performance in most scenarios, CTURNet still has some minor limitations. First, the model's computational complexity is relatively high. Although this does not affect most applications, it may pose challenges when handling extremely large datasets or real-time tasks. Second, while CTURNet performs well in most complex environments, segmentation accuracy may decrease in cases of extreme occlusion or very small targets particularly when rock fragments are tiny or fully overlapping. Finally, the model's training relies on a large amount of data. Nevertheless, these limitations do not significantly affect CTURNet's overall performance and instead provide clear directions for further research and optimization.

6. Conclusions and future work

This paper proposes a novel instance segmentation network, CTURNet, designed for the instance segmentation of post-blast open-pit rock fragments and subsequent measurement of rock pile size distribution. Compared with existing instance segmentation algorithms, CTURNet demonstrates significant advantages when handling rocks with fuzzy boundaries and complex shapes. To improve the identification of rocks with indistinct boundaries, we introduce a multi-stage contour enhancement module (MLCEM) that enhances rock contour features at multiple levels. We also design a Transformer-based FPN module, which, building on the original FPN, ensures long-range contextual dependencies across multiple scales and enhances adaptability in complex scenes. In addition, we design the UMask segmentation module, which enables precise capture of local texture details and improves rock segmentation accuracy. Through the synergistic effect of these modules, CTURNet is able to better address the challenges of overlapping rocks and blurred boundaries in instance segmentation tasks.

Training was conducted on both a self-built rock dataset and a public rock segmentation dataset. Experimental results show that CTURNet outperforms mainstream instance segmentation models. Compared with the baseline model, all evaluated metrics improved significantly AP increased by 3.29%, IoU by 4.34%, and Precision by 6.05%. Compared with SOTA models, CTURNet achieved improvements of at least 1.26%, 1.89%, 2.82%, and 1.95% in AP, IoU, Recall, and Precision, respectively. To further verify the segmentation accuracy of CTURNet, the equivalent rock sizes predicted by the network were compared with manually annotated measurements. The results indicate that the prediction errors for D10, D50, and D90 particle size ranges are all below 10%, demonstrating the model's high segmentation accuracy in practical applications. Although CTURNet exhibits superior performance in the current experiments, there remains room for further improvement. Future work could explore multimodal data fusion, such as integrating LiDAR or high-resolution imagery, to further enhance the model's robustness and accuracy in extremely complex scenarios. In addition, with the continued development of deep learning, automated feature selection and network architecture optimization may bring new breakthroughs. Future research could also investigate the use of transfer learning or limited labeled data to improve the model's generalization capability across diverse environments.

CRedit authorship contribution statement

Shiwei Wang: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Yu Liu:** Writing – review & editing, Supervision, Funding acquisition. **Jinguang Lin:** Investigation, Formal analysis, Conceptualization. **Jingyu Wang:** Formal analysis, Conceptualization. **Anxing Zhang:** Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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